

FIFA World Cup Analysis

Nguyen Dao Truong Vy

Student Information

Student Name: Nguyen Dao Truong Vy

Student Number: 22202382

Subject Code: COMP1013

Subject Name: Analytics Programming

GitHub Repository:

<https://github.com/yvette195/COMP1013-FIFA-WorldCup-Analysis>

Declaration

By including this statement, we the authors of this work, verify that:

- We hold a copy of this assignment that we can produce if the original is lost or damaged.
- We hereby certify that no part of this assignment/product has been copied from any other student work or from any other source except where due acknowledgement is made in the assignment.
- No part of this assignment/product has been written/produced for us by another person except where such collaboration has been authorised by the subject lecturer/tutor concerned.
- We are aware that this work may be reproduced and submitted to plagiarism detection software programs for the purpose of detecting possible plagiarism, which may retain a copy on its database for future plagiarism checking.
- We hereby certify that we have read and understand what the School of Computing, Engineering and Mathematics defines as minor and substantial breaches of misconduct as outlined in the learning guide for this unit.

Question 1: Data Inspection, Cleaning and Home Penalty Distribution

Rationale

I joined the four datasets because the information for this analysis was not stored in only one file. The `Matches` dataset contains the match results and scores, while the other datasets provide the team, stadium and tournament information. Joining them made it easier for me to use all the information in one data frame.

I changed "?" to NA because "?" is only a text value, but NA is recognised by R as a missing value. I changed `Result`, `Stage`, `Country` and `ExtraTime` to factors because these variables contain categories. I changed `HomePenalty` and `AwayPenalty` to numeric because they show the number of penalty goals.

I used a bar chart for `HomePenalty` because penalty scores are separate whole numbers. The chart makes it easy to see which penalty score appeared more often and which score appeared less often.

Load and Join the Data

```
# Read the data CSV files.
matches <- read_csv("../data/Matches.csv")
stadiums <- read_csv("../data/Stadiums.csv")
teams <- read_csv("../data/Teams.csv")
tournaments <- read_csv("../data/Tournaments.csv")

# Change every "?" value to NA.
matches[matches == "?"] <- NA
stadiums[stadiums == "?"] <- NA
teams[teams == "?"] <- NA
tournaments[tournaments == "?"] <- NA

# Join team information for the home team and away team.
df <- matches %>%
  left_join(
    teams %>%
      rename(
        HomeTeamID = TeamID,
        HomeTeamName = TeamName,
        HomeTeamCode = TeamCode
      ),
    by = "HomeTeamID"
  ) %>%
  left_join(
    teams %>%
      rename(
        AwayTeamID = TeamID,
        AwayTeamName = TeamName,
        AwayTeamCode = TeamCode
      ),
    by = "AwayTeamID"
  ) %>%
```

```
left_join(stadiums, by = "StadiumID") %>%
left_join(tournaments, by = "TournamentID")
```

Inspect and Clean the Data

```
# Display the first six rows.
head(df)
```

```
## # A tibble: 6 x 24
##   MatchID TournamentID MatchName Stage MatchDate MatchTime StadiumID HomeTeamID
##   <chr>      <chr>      <chr>    <chr> <chr>      <time>      <chr>      <chr>
## 1 M-1930-- WC-1930      France v~ grou~ 7/13/1930 15:00      S-193      T-28
## 2 M-1930-- WC-1930      United S~ grou~ 7/13/1930 15:00      S-192      T-80
## 3 M-1930-- WC-1930      Yugoslav~ grou~ 7/14/1930 12:45      S-192      T-84
## 4 M-1930-- WC-1930      Romania ~ grou~ 7/14/1930 14:50      S-193      T-59
## 5 M-1930-- WC-1930      Argentin~ grou~ 7/15/1930 16:00      S-192      T-03
## 6 M-1930-- WC-1930      Chile v ~ grou~ 7/16/1930 14:45      S-192      T-13
## # i 16 more variables: AwayTeamID <chr>, HomeTeamScore <dbl>,
## #   AwayTeamScore <dbl>, ExtraTime <dbl>, PenaltyShootout <dbl>,
## #   HomePenalty <chr>, AwayPenalty <chr>, Result <chr>, HomeTeamName <chr>,
## #   HomeTeamCode <chr>, AwayTeamName <chr>, AwayTeamCode <chr>,
## #   StadiumName <chr>, City <chr>, Country <chr>, TournamentName <chr>
```

```
# Check the number of rows and columns.
dim(df)
```

```
## [1] 964 24
```

```
# Check the variable types.
str(df)
```

```
## spc_tbl_ [964 x 24] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
## $ MatchID      : chr [1:964] "M-1930-01" "M-1930-02" "M-1930-03" "M-1930-04" ...
## $ TournamentID : chr [1:964] "WC-1930" "WC-1930" "WC-1930" "WC-1930" ...
## $ MatchName     : chr [1:964] "France v Mexico" "United States v Belgium" "Yugoslavia v Brazil" "R
## $ Stage         : chr [1:964] "group stage" "group stage" "group stage" "group stage" ...
## $ MatchDate     : chr [1:964] "7/13/1930" "7/13/1930" "7/14/1930" "7/14/1930" ...
## $ MatchTime     : 'hms' num [1:964] 15:00:00 15:00:00 12:45:00 14:50:00 ...
## ..- attr(*, "units")= chr "secs"
## $ StadiumID     : chr [1:964] "S-193" "S-192" "S-192" "S-193" ...
## $ HomeTeamID    : chr [1:964] "T-28" "T-80" "T-84" "T-59" ...
## $ AwayTeamID    : chr [1:964] "T-44" "T-06" "T-09" "T-54" ...
## $ HomeTeamScore : num [1:964] 4 3 2 3 1 3 4 3 1 1 ...
## $ AwayTeamScore : num [1:964] 1 0 1 1 0 0 0 0 0 0 ...
## $ ExtraTime     : num [1:964] 0 0 0 0 0 0 0 0 0 0 ...
## $ PenaltyShootout: num [1:964] 0 0 0 0 0 0 0 0 0 0 ...
## $ HomePenalty   : chr [1:964] NA "0" "0" "0" ...
## $ AwayPenalty   : chr [1:964] "0" "0" "0" NA ...
## $ Result        : chr [1:964] "home team win" "home team win" "home team win" "home team win" ...
## $ HomeTeamName  : chr [1:964] "France" "United States" "Yugoslavia" "Romania" ...
```

```
## $ HomeTeamCode : chr [1:964] "FRA" "USA" "YUG" "ROU" ...
## $ AwayTeamName : chr [1:964] "Mexico" "Belgium" "Brazil" "Peru" ...
## $ AwayTeamCode : chr [1:964] "MEX" "BEL" "BRA" "PER" ...
## $ StadiumName : chr [1:964] "Estadio Pocitos" "Estadio Gran Parque Central" "Estadio Gran Parque
## $ City : chr [1:964] "Montevideo" "Montevideo" "Montevideo" "Montevideo" ...
## $ Country : chr [1:964] "Uruguay" "Uruguay" "Uruguay" "Uruguay" ...
## $ TournamentName : chr [1:964] "1930 FIFA World Cup" "1930 FIFA World Cup" "1930 FIFA World Cup" "1
## - attr(*, "spec")=
## .. cols(
## .. MatchID = col_character(),
## .. TournamentID = col_character(),
## .. MatchName = col_character(),
## .. Stage = col_character(),
## .. MatchDate = col_character(),
## .. MatchTime = col_time(format = ""),
## .. StadiumID = col_character(),
## .. HomeTeamID = col_character(),
## .. AwayTeamID = col_character(),
## .. HomeTeamScore = col_double(),
## .. AwayTeamScore = col_double(),
## .. ExtraTime = col_double(),
## .. PenaltyShootout = col_double(),
## .. HomePenalty = col_character(),
## .. AwayPenalty = col_character(),
## .. Result = col_character()
## .. )
## - attr(*, "problems")=<externalptr>
```

Display a summary of the main variables.

```
summary(
  df[c(
    "Result",
    "Stage",
    "Country",
    "ExtraTime",
    "PenaltyShootout",
    "HomePenalty",
    "AwayPenalty"
  )]
)
```

##	Result	Stage	Country	ExtraTime
##	Length:964	Length:964	Length:964	Min. :0.00000
##	Class :character	Class :character	Class :character	1st Qu.:0.00000
##	Mode :character	Mode :character	Mode :character	Median :0.00000
##				Mean :0.07573
##				3rd Qu.:0.00000
##				Max. :1.00000
##	PenaltyShootout	HomePenalty	AwayPenalty	
##	Min. :0.00000	Length:964	Length:964	
##	1st Qu.:0.00000	Class :character	Class :character	
##	Median :0.00000	Mode :character	Mode :character	
##	Mean :0.03631			
##	3rd Qu.:0.00000			

```
## Max. :1.00000
```

```
# Convert categorical variables to factors.
df$Result <- as.factor(df$Result)
df$Stage <- as.factor(df$Stage)
df$Country <- as.factor(df$Country)
df$ExtraTime <- as.factor(df$ExtraTime)

# Convert penalty variables to numeric.
df$HomePenalty <- as.numeric(df$HomePenalty)
df$AwayPenalty <- as.numeric(df$AwayPenalty)

# Replace missing penalty values with zero.
df$HomePenalty[is.na(df$HomePenalty)] <- 0
df$AwayPenalty[is.na(df$AwayPenalty)] <- 0
```

Code Testing

```
# The number of rows should be unchanged after joining.
nrow(matches)
```

```
## [1] 964
```

```
nrow(df)
```

```
## [1] 964
```

```
# Check the classes after conversion.
class(df$Result)
```

```
## [1] "factor"
```

```
class(df$Stage)
```

```
## [1] "factor"
```

```
class(df$Country)
```

```
## [1] "factor"
```

```
class(df$ExtraTime)
```

```
## [1] "factor"
```

```
class(df$HomePenalty)
```

```
## [1] "numeric"
```

```
class(df$AwayPenalty)
```

```
## [1] "numeric"
```

```
# Check that no missing penalty values remain.
```

```
sum(is.na(df$HomePenalty))
```

```
## [1] 0
```

```
sum(is.na(df$AwayPenalty))
```

```
## [1] 0
```

The number of rows in `matches` and `df` should be the same. This is important because joining the other datasets should add more columns, but it should not remove or add match records.

The four category variables should return `"factor"`. The penalty variables should return `"numeric"`. The two missing-value checks should return zero. These checks help me confirm that the data was changed into the correct format before doing the analysis.

Results and Interpretation

```
# Keep only matches involving a penalty shootout.
```

```
penalty_matches <- df %>%  
  filter(PenaltyShootout == 1)
```

```
# Create a frequency table for HomePenalty.
```

```
penalty_table <- table(penalty_matches$HomePenalty)  
penalty_table
```

```
##
```

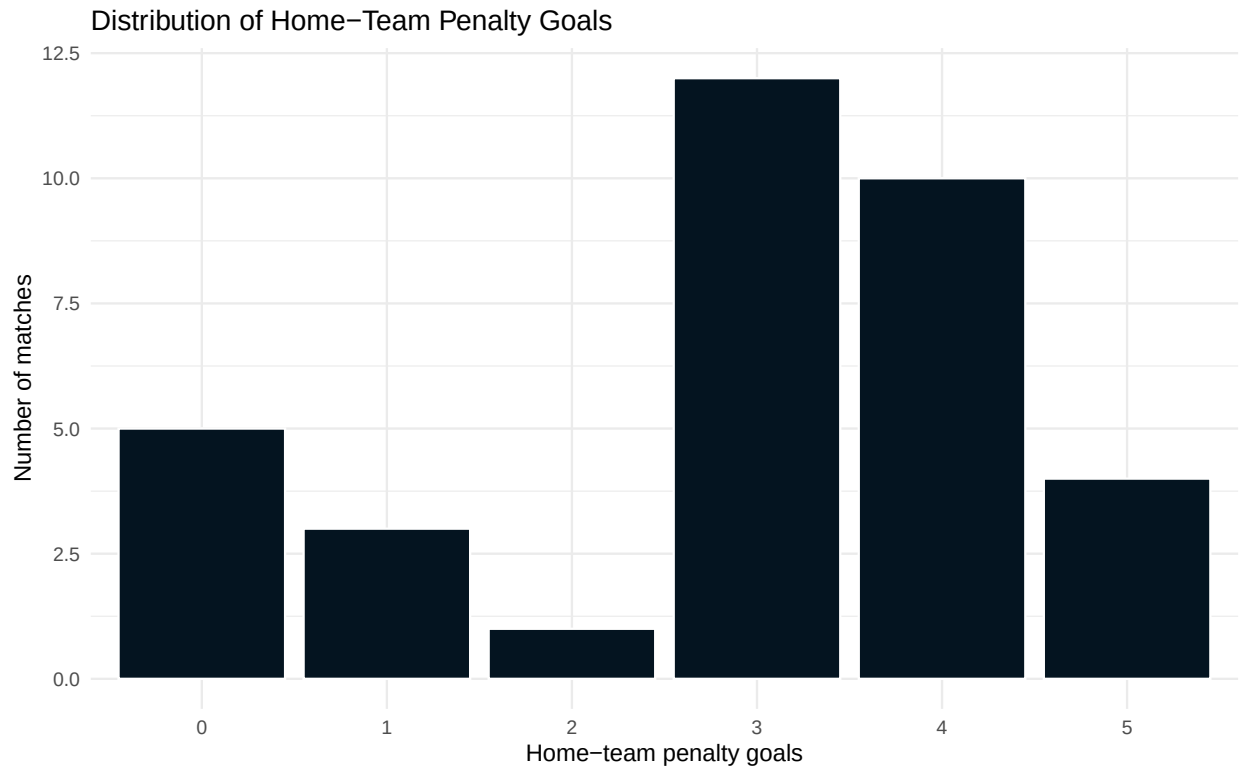
```
## 0  1  2  3  4  5
```

```
## 5  3  1 12 10  4
```

```
# Find the most common HomePenalty value.
```

```
most_common_penalty <- names(penalty_table)[which.max(penalty_table)]  
most_common_count <- max(penalty_table)
```

```
ggplot(penalty_matches, aes(x = factor(HomePenalty))) +  
  geom_bar(  
    fill = darkest_colour,  
    colour = "white"  
  ) +  
  labs(  
    title = "Distribution of Home-Team Penalty Goals",  
    x = "Home-team penalty goals",  
    y = "Number of matches"  
  ) +  
  theme_minimal()
```



There were 35 matches that included a penalty shootout. The most common value of `HomePenalty` was 3. This value appeared in 12 matches.

From the chart, some penalty scores appeared more often than the other scores. The taller bars mean that the score happened in more matches. The shorter bars mean that the score was less common. The distribution was not equal because the bars did not have the same height.

Question 2: Distribution of Home Team Scores

Rationale

I used histograms because `HomeTeamScore` is a numerical variable. Football scores are whole numbers, so I used a bin width of one. This means that each bar represents one score, such as 0, 1, 2 or 3 goals.

I compared the home-team scores by `Stage`, `TournamentName` and `AwayScoreGroup`. I separated these comparisons because putting all the information in one chart would make the chart too crowded and difficult to read.

I also used `aggregate()` to calculate the mean and median for each group. The histograms show the shape of the score distribution, while the mean and median give simple numbers that can be compared between groups.

Distribution by Competition Stage

```
# Calculate the mean and median HomeTeamScore for each stage.
stage_mean <- aggregate(
  HomeTeamScore ~ Stage,
  mean,
  data = df
)

stage_median <- aggregate(
  HomeTeamScore ~ Stage,
  median,
  data = df
)

stage_mean
```

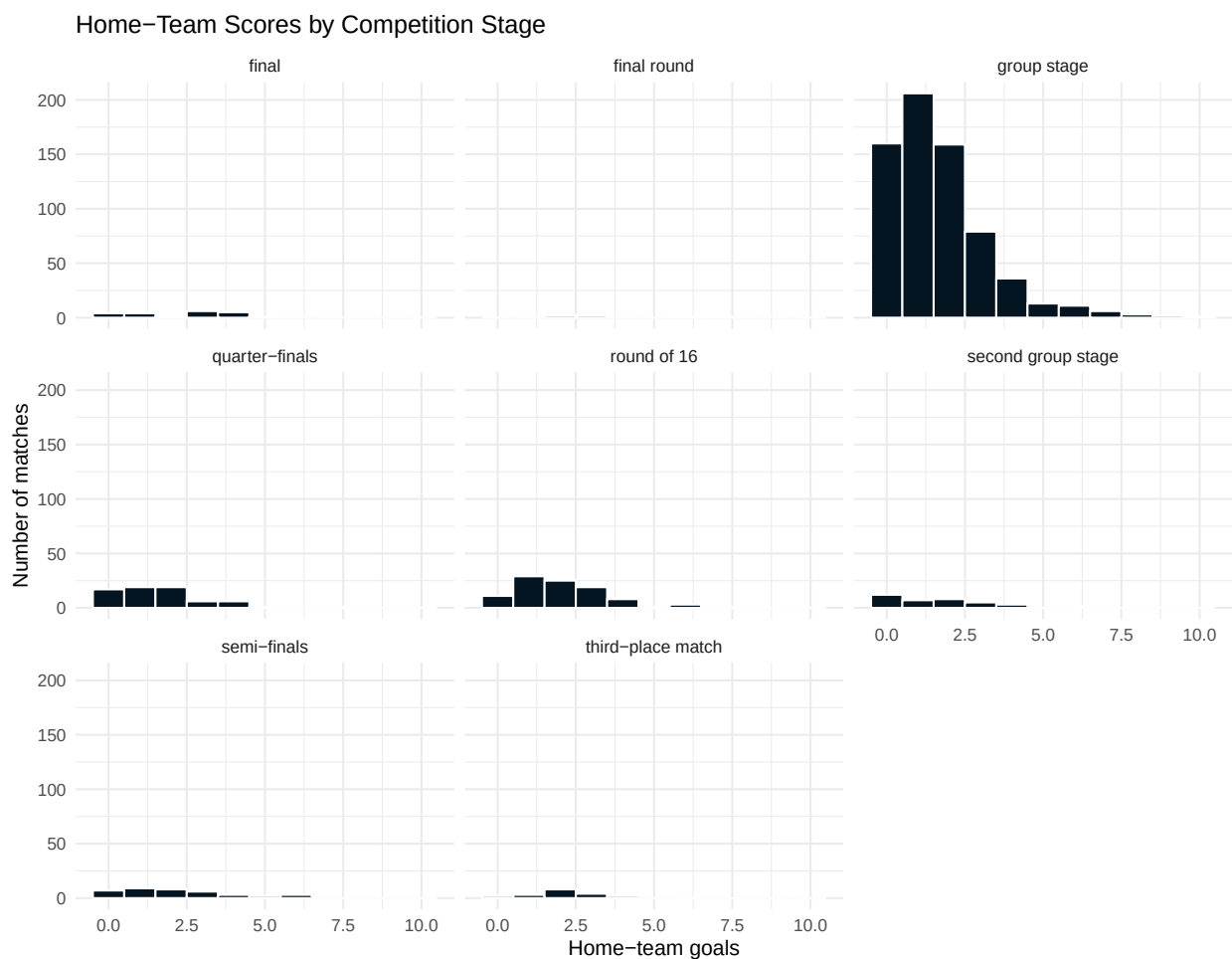
```
##           Stage HomeTeamScore
## 1           final      2.333333
## 2    final round      3.833333
## 3    group stage      1.671598
## 4 quarter-finals      1.700000
## 5    round of 16      2.041237
## 6 second group stage      1.555556
## 7    semi-finals      2.184211
## 8 third-place match      2.250000
```

```
stage_median
```

```
##           Stage HomeTeamScore
## 1           final              3
## 2    final round              3
## 3    group stage              1
## 4 quarter-finals              1
## 5    round of 16              2
## 6 second group stage          1
## 7    semi-finals              2
## 8 third-place match          2
```



```
ggplot(df, aes(x = HomeTeamScore)) +
  geom_histogram(
    binwidth = 1,
    fill = darkest_colour,
    colour = "white"
  ) +
  facet_wrap(~ Stage) +
  labs(
    title = "Home-Team Scores by Competition Stage",
    x = "Home-team goals",
    y = "Number of matches"
  ) +
  theme_minimal()
```



Distribution by Tournament Name

```
# Calculate the mean and median HomeTeamScore for each tournament.
tournament_mean <- aggregate(
  HomeTeamScore ~ TournamentName,
  mean,
```

```

data = df
)

tournament_median <- aggregate(
  HomeTeamScore ~ TournamentName,
  median,
  data = df
)

tournament_mean

```

```

##      TournamentName HomeTeamScore
## 1  1930 FIFA World Cup      3.277778
## 2  1934 FIFA World Cup      2.823529
## 3  1938 FIFA World Cup      3.388889
## 4  1950 FIFA World Cup      3.136364
## 5  1954 FIFA World Cup      4.192308
## 6  1958 FIFA World Cup      2.514286
## 7  1962 FIFA World Cup      2.156250
## 8  1966 FIFA World Cup      2.156250
## 9  1970 FIFA World Cup      2.218750
## 10 1974 FIFA World Cup      1.342105
## 11 1978 FIFA World Cup      1.894737
## 12 1982 FIFA World Cup      1.846154
## 13 1986 FIFA World Cup      1.384615
## 14 1990 FIFA World Cup      1.230769
## 15 1994 FIFA World Cup      1.596154
## 16 1998 FIFA World Cup      1.531250
## 17 2002 FIFA World Cup      1.390625
## 18 2006 FIFA World Cup      1.343750
## 19 2010 FIFA World Cup      1.187500
## 20 2014 FIFA World Cup      1.265625
## 21 2018 FIFA World Cup      1.421875
## 22 2022 FIFA World Cup      1.578125

```

```

tournament_median

```

```

##      TournamentName HomeTeamScore
## 1  1930 FIFA World Cup          3.0
## 2  1934 FIFA World Cup          3.0
## 3  1938 FIFA World Cup          3.0
## 4  1950 FIFA World Cup          2.5
## 5  1954 FIFA World Cup          4.0
## 6  1958 FIFA World Cup          2.0
## 7  1962 FIFA World Cup          2.0
## 8  1966 FIFA World Cup          2.0
## 9  1970 FIFA World Cup          2.0
## 10 1974 FIFA World Cup          1.0
## 11 1978 FIFA World Cup          2.0
## 12 1982 FIFA World Cup          1.0
## 13 1986 FIFA World Cup          1.0
## 14 1990 FIFA World Cup          1.0
## 15 1994 FIFA World Cup          1.0

```

```
## 16 1998 FIFA World Cup      1.0
## 17 2002 FIFA World Cup      1.0
## 18 2006 FIFA World Cup      1.0
## 19 2010 FIFA World Cup      1.0
## 20 2014 FIFA World Cup      1.0
## 21 2018 FIFA World Cup      1.0
## 22 2022 FIFA World Cup      1.0
```

```
ggplot(df, aes(x = HomeTeamScore)) +
  geom_histogram(
    binwidth = 1,
    fill = darkest_colour,
    colour = "white"
  ) +
  facet_wrap(~ TournamentName) +
  labs(
    title = "Home-Team Scores by Tournament",
    x = "Home-team goals",
    y = "Number of matches"
  ) +
  theme_minimal()
```

Home-Team Scores by Tournament



Distribution by Away-Team Score Group

```
# Create an empty variable for the away-team score groups.
df$AwayScoreGroup <- NA
```

```

# Assign each match to one of the three required groups.
df$AwayScoreGroup[df$AwayTeamScore <= 1] <- "0-1 goals"

df$AwayScoreGroup[
  df$AwayTeamScore >= 2 &
  df$AwayTeamScore <= 3
] <- "2-3 goals"

df$AwayScoreGroup[df$AwayTeamScore >= 4] <- "4 or more goals"

# Convert the new variable to a factor.
df$AwayScoreGroup <- as.factor(df$AwayScoreGroup)

# Check the number of matches in each group.
table(df$AwayScoreGroup)

```

```

##
##      0-1 goals      2-3 goals 4 or more goals
##           701           237           26

```

```

# Count the number of matches in each group.
away_group_counts <- table(df$AwayScoreGroup)

# Put the groups in order from the smallest count to the largest count.
away_group_order <- names(sort(away_group_counts))

# Give a light colour to the smallest group and a dark colour to the largest group.
away_group_colours <- setNames(
  c(
    lightest_colour,
    middle_colour,
    darkest_colour
  ),
  away_group_order
)

# Calculate the mean and median HomeTeamScore for each group.
away_group_mean <- aggregate(
  HomeTeamScore ~ AwayScoreGroup,
  mean,
  data = df
)

away_group_median <- aggregate(
  HomeTeamScore ~ AwayScoreGroup,
  median,
  data = df
)

away_group_mean

```

```

##      AwayScoreGroup HomeTeamScore

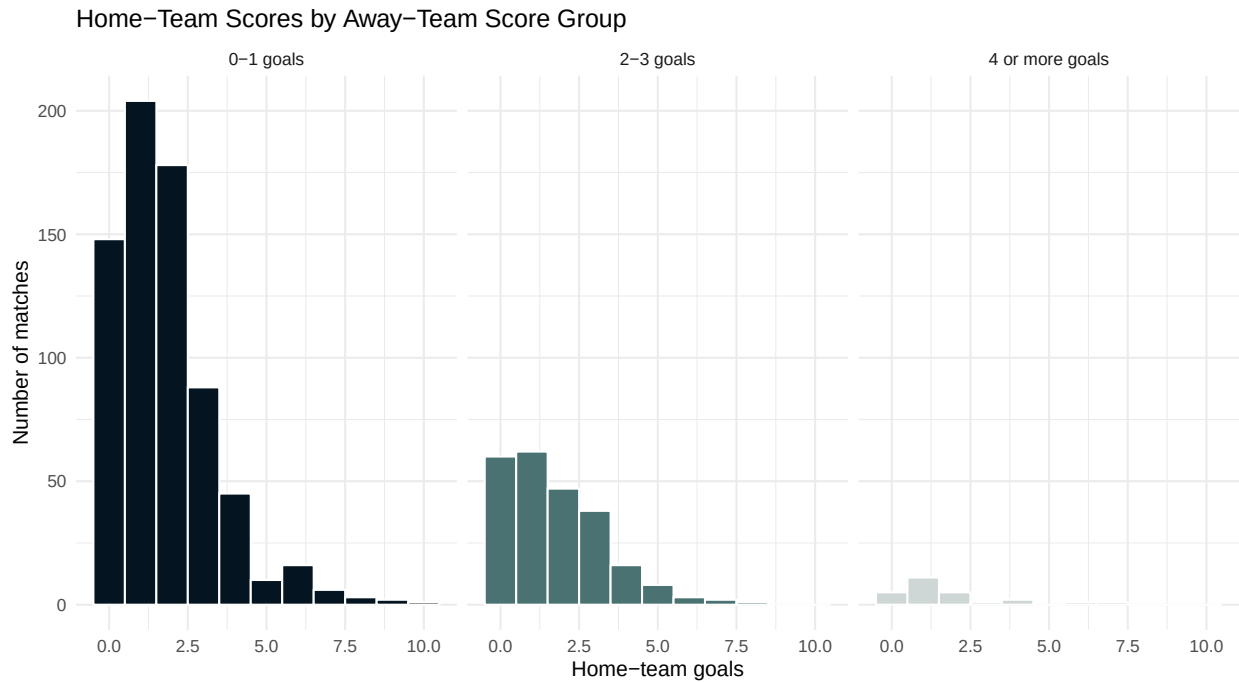
```

```
## 1      0-1 goals      1.774608
## 2      2-3 goals      1.746835
## 3 4 or more goals      1.730769
```

```
away_group_median
```

```
##      AwayScoreGroup HomeTeamScore
## 1      0-1 goals      1
## 2      2-3 goals      1
## 3 4 or more goals      1
```

```
ggplot(
  df,
  aes(
    x = HomeTeamScore,
    fill = AwayScoreGroup
  )
) +
  geom_histogram(
    binwidth = 1,
    colour = "white"
  ) +
  facet_wrap(~ AwayScoreGroup) +
  scale_fill_manual(
    values = away_group_colours
  ) +
  guides(fill = "none") +
  labs(
    title = "Home-Team Scores by Away-Team Score Group",
    x = "Home-team goals",
    y = "Number of matches"
  ) +
  theme_minimal()
```



Code Testing

```
# Check the score variable types.
class(df$HomeTeamScore)
```

```
## [1] "numeric"
```

```
class(df$AwayTeamScore)
```

```
## [1] "numeric"
```

```
# Check for missing score values.
sum(is.na(df$HomeTeamScore))
```

```
## [1] 0
```

```
sum(is.na(df$AwayTeamScore))
```

```
## [1] 0
```

```
# Check the new group variable.
class(df$AwayScoreGroup)
```

```
## [1] "factor"
```

```
table(df$AwayScoreGroup)
```

```
##
##      0-1 goals      2-3 goals 4 or more goals
##           701           237           26
```

```
sum(is.na(df$AwayScoreGroup))
```

```
## [1] 0
```

```
# Check that all matches are included in the three groups.
sum(table(df$AwayScoreGroup))
```

```
## [1] 964
```

```
nrow(df)
```

```
## [1] 964
```

HomeTeamScore and AwayTeamScore should be numeric variables. R may show "integer" because football scores are whole numbers, and this is still suitable for this analysis.

The missing-value results should be zero. AwayScoreGroup should be a factor and it should not contain missing values. The total number of matches in the three groups should also be equal to the number of rows in df. This check confirms that every match was placed into one group.

Results and Interpretation

The stage with the highest average home-team score was **final round**, with an average of 3.83 goals. The stage with the lowest average was **second group stage**, with an average of 1.56 goals.

The histograms show that most home-team scores were around the lower score values. Some stages had more matches than the other stages, so their bars were also taller. The chart is useful for comparing the general score pattern, but the mean is also needed because the number of matches was different between stages.

The tournament with the highest average home-team score was **1954 FIFA World Cup**, with 4.19 goals. The tournament with the lowest average was **2010 FIFA World Cup**, with 1.19 goals.

The score distribution was not exactly the same for every tournament. Some tournaments had more high-scoring matches, while some tournaments had more scores around zero, one or two goals. However, the number of matches was also different between tournaments, so the charts should be compared together with the mean and median.

For the away-team score groups, **0-1 goals** had the highest average home-team score, at 1.77 goals. This shows that the home-team score was different between the three away-score groups. However, this is only a relationship found in the dataset. It does not mean that the away-team score directly caused the home-team score.

Question 3: Penalty Shootout Participation

Rationale

I first selected only the matches where `PenaltyShootout` was equal to 1. I did this because Question 3 only asks about matches that included a penalty shootout.

After filtering the matches, I grouped the data by the team name and team code. I counted the number of matches for each team by using `summarise()` and `n()`.

I did the home-team and away-team analysis separately because a team may be recorded as the home team in one match and the away team in another match. I then sorted the results from the highest number to the lowest number. Finally, I used `head()` to select the first five teams.

Filter Penalty Shootout Matches

```
# Keep only matches that involved a penalty shootout.
penalty_matches <- df %>%
  filter(PenaltyShootout == 1)

# Check the number of penalty-shootout matches.
nrow(penalty_matches)
```

```
## [1] 35
```

```
# Check that the filtered data contains only value 1.
table(penalty_matches$PenaltyShootout)
```

```
##
##  1
## 35
```

Top Five Home Teams

```
# Group the penalty matches by home-team name and code.
# Count the number of matches for each home team.
home_penalty_count <- penalty_matches %>%
  group_by(HomeTeamName, HomeTeamCode) %>%
  summarise(NumberOfMatches = n())

# Sort the teams from the highest count to the lowest count.
home_penalty_count <- home_penalty_count %>%
  arrange(desc(NumberOfMatches))

# Select the first five teams.
top5_home <- head(home_penalty_count, 5)

# Change the column names to match the assignment requirement.
names(top5_home) <- c(
```

```

  "TeamName",
  "TeamCode",
  "NumberOfMatches"
)

# Display the top five home teams as a table.
top5_home

```

```

## # A tibble: 5 x 3
## # Groups:   TeamName [5]
##   TeamName    TeamCode NumberOfMatches
##   <chr>      <chr>          <int>
## 1 Argentina    ARG              4
## 2 Brazil      BRA              4
## 3 Spain        ESP              4
## 4 Netherlands  NLD              3
## 5 West Germany DEU              3

```

Top Five Away Teams

```

# Group the penalty matches by away-team name and code.
# Count the number of matches for each away team.
away_penalty_count <- penalty_matches %>%
  group_by(AwayTeamName, AwayTeamCode) %>%
  summarise(NumberOfMatches = n())

# Sort the teams from the highest count to the lowest count.
away_penalty_count <- away_penalty_count %>%
  arrange(desc(NumberOfMatches))

# Select the first five teams.
top5_away <- head(away_penalty_count, 5)

# Change the column names to match the assignment requirement.
names(top5_away) <- c(
  "TeamName",
  "TeamCode",
  "NumberOfMatches"
)

# Display the top five away teams as a table.
top5_away

```

```

## # A tibble: 5 x 3
## # Groups:   TeamName [5]
##   TeamName    TeamCode NumberOfMatches
##   <chr>      <chr>          <int>
## 1 France     FRA              5
## 2 Argentina  ARG              3
## 3 England    ENG              3
## 4 Croatia    HRV              2
## 5 Italy       ITA              2

```

Code Testing

```
# Check that all filtered matches involved a penalty shootout.  
table(penalty_matches$PenaltyShootout)
```

```
##  
## 1  
## 35
```

```
# Both top-five tables should contain five rows.  
nrow(top5_home)
```

```
## [1] 5
```

```
nrow(top5_away)
```

```
## [1] 5
```

```
# Check that both tables have the required column names.  
names(top5_home)
```

```
## [1] "TeamName"      "TeamCode"      "NumberOfMatches"
```

```
names(top5_away)
```

```
## [1] "TeamName"      "TeamCode"      "NumberOfMatches"
```

```
# Check for missing values in team names and codes.  
sum(is.na(top5_home$TeamName))
```

```
## [1] 0
```

```
sum(is.na(top5_home$TeamCode))
```

```
## [1] 0
```

```
sum(is.na(top5_away$TeamName))
```

```
## [1] 0
```

```
sum(is.na(top5_away$TeamCode))
```

```
## [1] 0
```

```
# Check that the match counts are sorted from high to low.
top5_home$NumberOfMatches
```

```
## [1] 4 4 4 3 3
```

```
top5_away$NumberOfMatches
```

```
## [1] 5 3 3 2 2
```

The `PenaltyShootout` table should only show the value 1. This confirms that the filtered data only contains penalty-shootout matches.

Both final tables should contain five rows. They should also contain the columns `TeamName`, `TeamCode` and `NumberOfMatches`. The missing-value checks should return zero. The match counts should be shown from the highest value to the lowest value.

Results and Interpretation

The first team in the home-team table was **Argentina**. This team appeared in 4 penalty-shootout matches as the home team.

The first team in the away-team table was **France**. This team appeared in 5 penalty-shootout matches as the away team.

There were 1 teams that appeared in both top-five tables. These teams were **Argentina**. The two rankings were **different** when I compared the team names and their order.

The home and away rankings can be different because the home-team and away-team labels can change from one match to another match. In this dataset, being called the home team also does not always mean that the team played in its own country.

Question 4: Factors Influencing Match Outcomes

Selected Factors and Rationale

I selected `Stage` and `ExtraTime` as the two factors.

I selected `Stage` because matches may be played differently at different parts of the tournament. For example, a group-stage match may not have the same situation as a semi-final or final match. Teams may use a different strategy because the importance of the match is different.

I selected `ExtraTime` because matches that continue after regular time may have a different result pattern. Extra time may happen when the teams are close in performance, so I wanted to compare these matches with the other matches.

I used frequency tables first because they show the number of home wins, away wins and draws. I also used row percentages because the number of matches was not the same in every group. The proportional bar charts make the three result categories easier to compare.

Overall Match Results

```
# Count the number of matches in each Result category.
overall_result_table <- table(df$Result)
```

```
overall_result_table
```

```
##
## away team win      draw home team win
##           240           179           545
```

```
# Put the Result categories in order
# from the smallest count to the largest count.
result_order <- names(sort(overall_result_table))
```

```
# Give a light colour to the least common result
# and a dark colour to the most common result.
```

```
result_colours <- setNames(
  c(
    lightest_colour,
    middle_colour,
    darkest_colour
  ),
  result_order
)
```

```
# Find the most common match result.
```

```
most_common_result <- names(overall_result_table)[
  which.max(overall_result_table)
]
```

```
most_common_result_count <- max(overall_result_table)
```

Factor 1: Competition Stage

```
# Create a table of Stage and Result.
```

```
stage_result_count <- table(  
  df$Stage,  
  df$Result  
)
```

```
stage_result_count
```

```
##  
##               away team win draw home team win  
##   final               4     0             17  
##   final round          0     1             5  
##   group stage        164   166           346  
##   quarter-finals      24     2            44  
##   round of 16         21     2            74  
##   second group stage  11     8            17  
##   semi-finals        10     0            28  
##   third-place match   6     0            14
```

```
# Convert the counts into row percentages.
```

```
# margin = 1 calculates percentages within each Stage.
```

```
stage_result_percent <- prop.table(  
  stage_result_count,  
  margin = 1  
) * 100
```

```
# Round the percentages to one decimal place.
```

```
stage_result_percent <- round(  
  stage_result_percent,  
  1  
)
```

```
stage_result_percent
```

```
##  
##               away team win draw home team win  
##   final               19.0  0.0             81.0  
##   final round          0.0 16.7             83.3  
##   group stage        24.3 24.6             51.2  
##   quarter-finals      34.3  2.9             62.9  
##   round of 16         21.6  2.1             76.3  
##   second group stage  30.6 22.2             47.2  
##   semi-finals        26.3  0.0             73.7  
##   third-place match   30.0  0.0             70.0
```

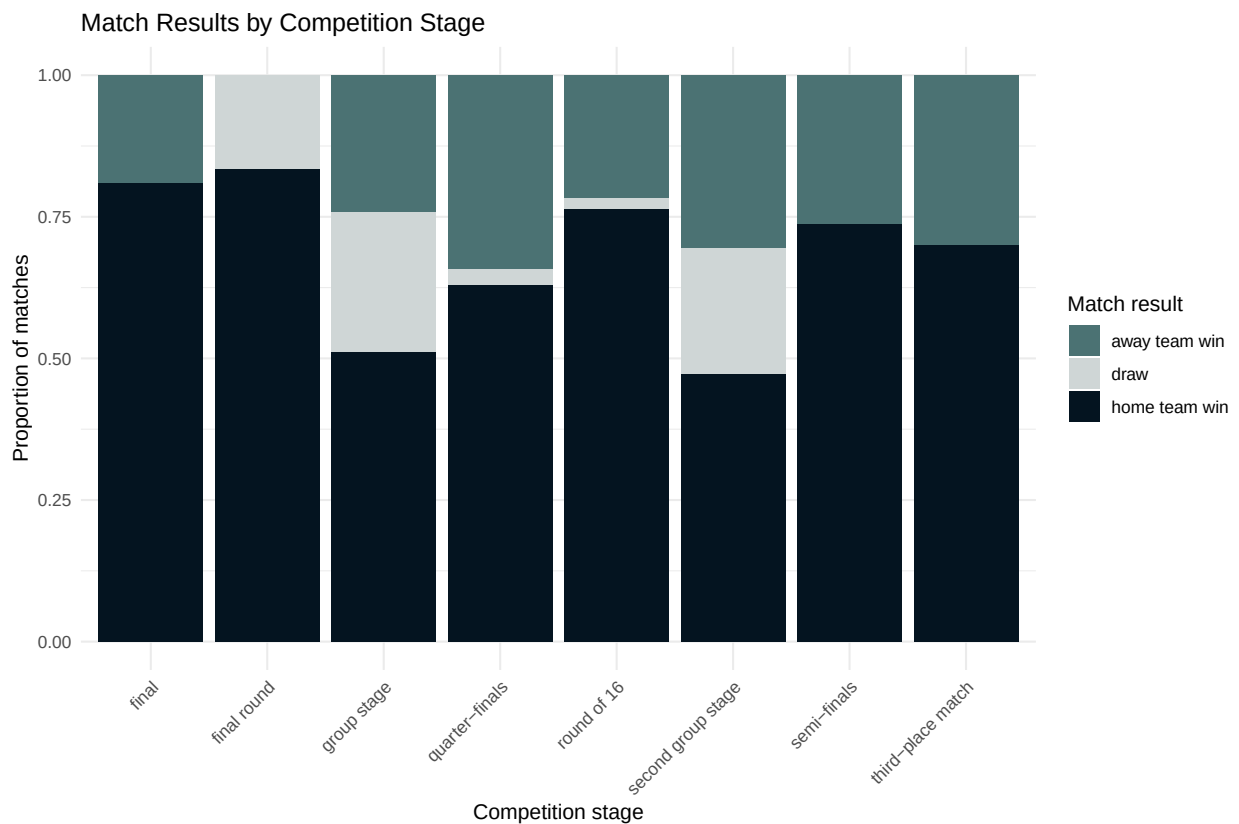
```
# Create a proportional bar chart of Result for each Stage.
```

```
ggplot(  
  df,  
  aes(  
    x = Stage,
```

```

    fill = Result
  )
) +
  geom_bar(position = "fill") +
  scale_fill_manual(
    values = result_colours
  ) +
  labs(
    title = "Match Results by Competition Stage",
    x = "Competition stage",
    y = "Proportion of matches",
    fill = "Match result"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(
      angle = 45,
      hjust = 1
    )
  )
)

```



Factor 2: Extra Time

```
# Create a table of ExtraTime and Result.
```

```
extra_time_result_count <- table(
  df$ExtraTime,
  df$Result
)
```

```
extra_time_result_count
```

```
##
```

```
##      away team win draw home team win
```

```
##    0          215  173          503
```

```
##    1           25    6           42
```

```
# Convert the counts into row percentages.
```

```
# margin = 1 calculates percentages within each ExtraTime group.
```

```
extra_time_result_percent <- prop.table(
  extra_time_result_count,
  margin = 1
) * 100
```

```
# Round the percentages to one decimal place.
```

```
extra_time_result_percent <- round(
  extra_time_result_percent,
  1
)
```

```
extra_time_result_percent
```

```
##
```

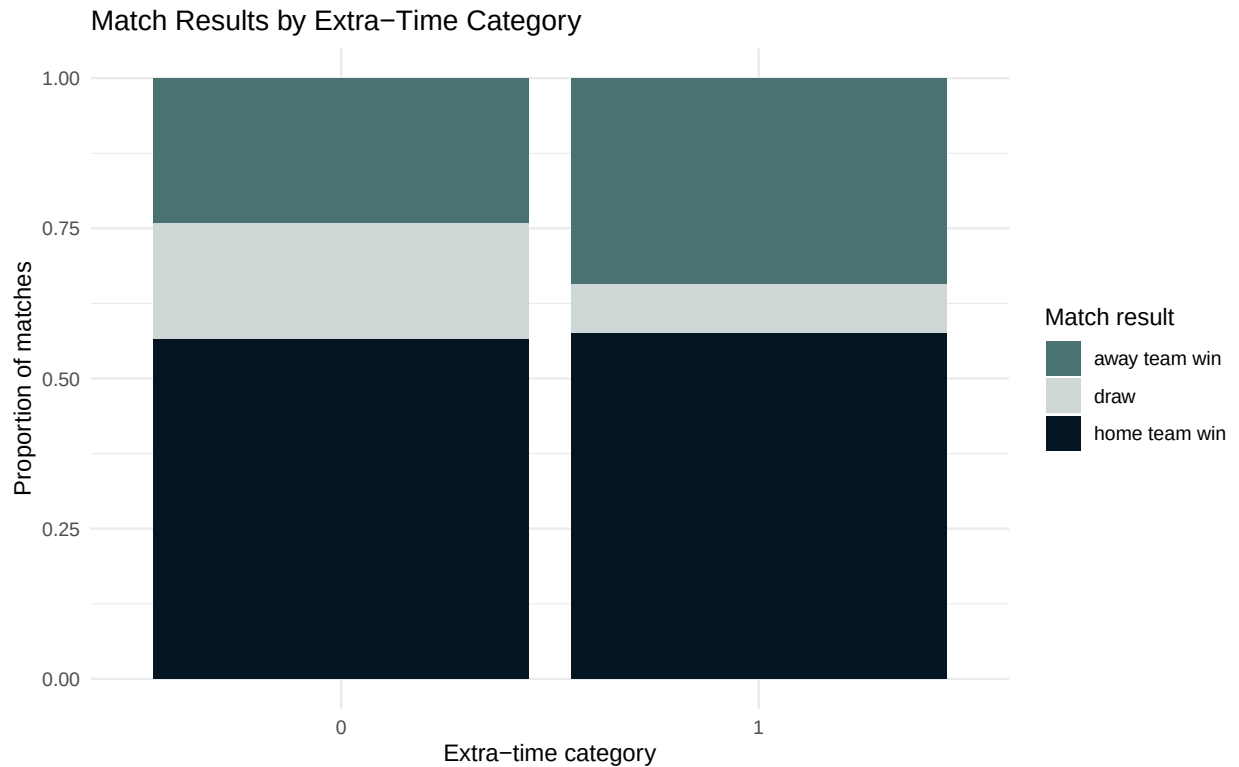
```
##      away team win draw home team win
```

```
##    0          24.1 19.4          56.5
```

```
##    1          34.2  8.2          57.5
```

```
# Create a proportional bar chart of Result by ExtraTime.
```

```
ggplot(
  df,
  aes(
    x = ExtraTime,
    fill = Result
  )
) +
  geom_bar(position = "fill") +
  scale_fill_manual(
    values = result_colours
  ) +
  labs(
    title = "Match Results by Extra-Time Category",
    x = "Extra-time category",
    y = "Proportion of matches",
    fill = "Match result"
  ) +
  theme_minimal()
```

Code Testing

```
# Check that the three variables are factors.
```

```
class(df$Result)
```

```
## [1] "factor"
```

```
class(df$Stage)
```

```
## [1] "factor"
```

```
class(df$ExtraTime)
```

```
## [1] "factor"
```

```
# Display the categories in each variable.
```

```
levels(df$Result)
```

```
## [1] "away team win" "draw"          "home team win"
```

```
levels(df$Stage)
```

```
## [1] "final"          "final round"    "group stage"
## [4] "quarter-finals" "round of 16"    "second group stage"
## [7] "semi-finals"    "third-place match"
```

```
levels(df$ExtraTime)
```

```
## [1] "0" "1"
```

```
# Check for missing values.
```

```
sum(is.na(df$Result))
```

```
## [1] 0
```

```
sum(is.na(df$Stage))
```

```
## [1] 0
```

```
sum(is.na(df$ExtraTime))
```

```
## [1] 0
```

```
# The total number in each table should equal the number of rows in df.
```

```
sum(stage_result_count)
```

```
## [1] 964
```

```
nrow(df)
```

```
## [1] 964
```

```
sum(extra_time_result_count)
```

```
## [1] 964
```

```
nrow(df)
```

```
## [1] 964
```

```
# Each row of the percentage tables should total approximately 100%.
```

```
rowSums(stage_result_percent)
```

```
##           final           final round           group stage           quarter-finals
##           100.0           100.0           100.1           100.1
## round of 16 second group stage           semi-finals third-place match
##           100.0           100.0           100.0           100.0
```

```
rowSums(extra_time_result_percent)
```

```
##      0      1
## 100.0  99.9
```

Result, Stage and ExtraTime should return "factor". Their levels show the categories stored inside each variable.

The missing-value checks should return zero. The total value from each count table should be equal to the number of rows in df. The rows in the percentage tables should add to around 100%. A result such as 99.9% or 100.1% can happen because the percentages were rounded.

Results and Interpretation

For all matches, the most common result was **home team win**. This result happened in 545 matches.

For the competition stages, **final round** had the highest percentage of home-team wins, at 83.3%. The percentage table shows that the result pattern was not exactly the same for every stage. Some stages had a higher percentage of home-team wins, while some stages had more away-team wins or draws.

For extra time, the category **1** had the highest percentage of home-team wins, at 57.5%. The chart also shows that the result distribution was different between the extra-time categories.

These results show that **Stage** and **ExtraTime** have a relationship with match results in this dataset. However, the analysis does not prove that these two factors directly caused a home-team win, away-team win or draw. Other factors may also affect the match result.

Question 5: Git and GitHub

GitHub Repository

Git and GitHub were used to manage the development of this project.

The GitHub repository is available at:

<https://github.com/yvette195/COMP1013-FIFA-WorldCup-Analysis>

Git was used to record changes made to the project. GitHub was used to store the repository online and maintain a history of the project development.

Project Folder Structure

The project was organised into separate folders:

- **data/** contains the four FIFA World Cup datasets.
- **scripts/** contains the R scripts used for Questions 1 to 4.
- **report/** contains the R Markdown report.
- **figures/** contains images used in the report.
- **COMP1013-FIFA-WorldCup-Analysis.Rproj** is the RStudio project file.

This folder structure keeps the datasets, scripts, report and figures separate, making the project easier to understand and maintain.

Commit History

```
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013
-FIFA-WorldCup-Analysis$ git log --oneline --decorate -12
720475e (HEAD -> main, origin/main, origin/HEAD) Revise explanat
ions and update chart colours
b29a7c7 Complete Question 5 and final report
ab15faa Complete Questions 3 and 4 analysis
c30d58a Fix report paths and update Questions 1 and 2
d787645 Change path of datasets
b0b1751 Simplify code and complete Questions 1 and 2
5abea0e Complete Questions 1 and 2 data analysis
2006d5f Complete Question 1 data preparation and penalty analysi
s
cb5ebd3 Add FIFA World Cup datasets
c29a184 Set up RStudio project and report template
6cf6ab0 Create initial project structure
e273768 Initial commit
```

Explanation of the Commit History

The commit history shows that I developed the project step by step. I did not upload all the work in only one commit.

1. **Initial commit**
This was the first commit used to start the repository.
2. **Create initial project structure**
This commit created the main folders for the project, including the data, scripts, report and figures folders.
3. **Set up RStudio project and report template**
This commit added the RStudio project file and the first R Markdown report template.
4. **Add FIFA World Cup datasets**
This commit added the four CSV datasets used for the assignment.
5. **Complete Question 1 data preparation and penalty analysis**
This commit added the data cleaning and penalty analysis for Question 1.
6. **Complete Questions 1 and 2 data analysis**
This commit added the main analysis for Questions 1 and 2.
7. **Simplify code and complete Questions 1 and 2**
This commit changed some code into simpler code and completed the first two questions.
8. **Change path of datasets**
This commit changed the dataset paths so the files could be read correctly from the report folder.
9. **Fix report paths and update Questions 1 and 2**
This commit fixed more file-path problems and updated the content for Questions 1 and 2.
10. **Complete Questions 3 and 4 analysis**
This commit added the penalty-shootout rankings and the match-outcome analysis.
11. **Complete Question 5 and final report**
This commit added the GitHub section, the commit-history screenshot and the first final PDF report.
12. **Revise explanations and update chart colours**
This commit rewrote the rationale and explanation sections using simpler English. It also changed the charts to use the same colour palette.

The commit messages show the main progress of my assignment. They also make it easier to understand what I changed at each stage.